

Dataset Link:

<https://www.kaggle.com/datasets/sabahezaraki/breast-ultrasound-images-dataset>

Overview of the Dataset:

Images of breast ultrasounds taken from women aged 25 to 75 were included in the baseline dataset. This information was arranged in 2018. Six hundred women make up the dataset. 780 photos with an average pixel size of 500*500 make up the dataset. However the dataset contains in total 1578 images including Ultrasound and Masked images In PNG format. The original photos are displayed beside the ground truth photos. The pictures fall into three categories: benign, malignant, and normal. The dataset contains some Ultrasound images with two affected places and that makes 2 masked images

Creating a Dataframe:

I created a dataframe, by iterating the folder and displayed the first 5 images to see how does that look.

SL	class	filename	is_mask	image_id
0	malignant	malignant (1).png	False	malignant (1)
1	malignant	malignant (1)_mask.png	True	malignant (1)
2	malignant	malignant (10).png	False	malignant (10)
3	malignant	malignant (10)_mask.png	True	malignant (10)
4	malignant	malignant (100).png	False	malignant (100)

Data Cleaning:

Now, to clean the data first we merge those masked images that contain two affected places, so that they become one and use that image to train our model. The figure below shows, what was done:



Furthermore, as all the images are not of same size, I resized them to 256x256.

Dataset Metadata

Example of metadata of one image:

Image ID:

Dimension: 562x471

Width: 562 px

Height: 471 px

Horizontal Resolution: 96 DPI

Vertical Resolution: 96 DPI

Bit Depth: 24

File:

Item type: PNG File

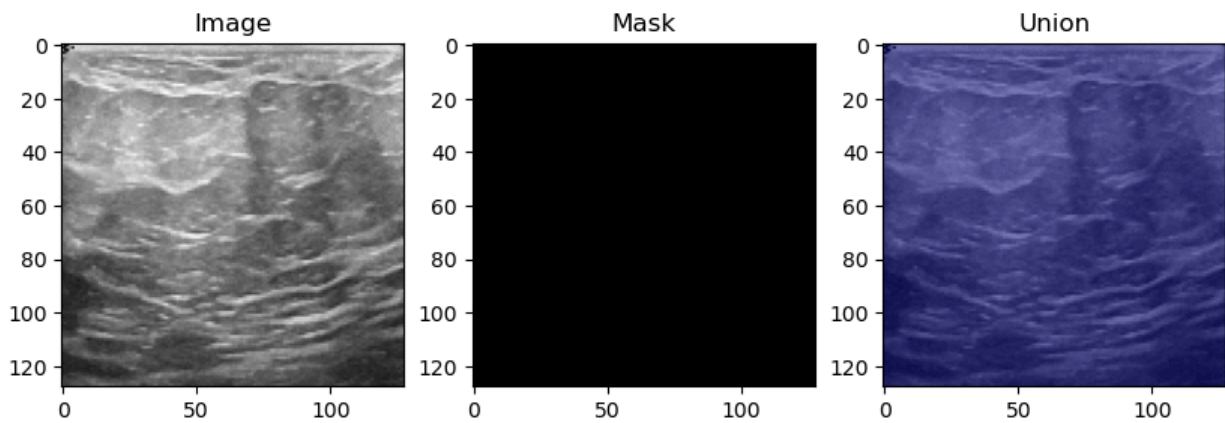
Size: 326 KB

Attributes: A

Sample Images from each class:

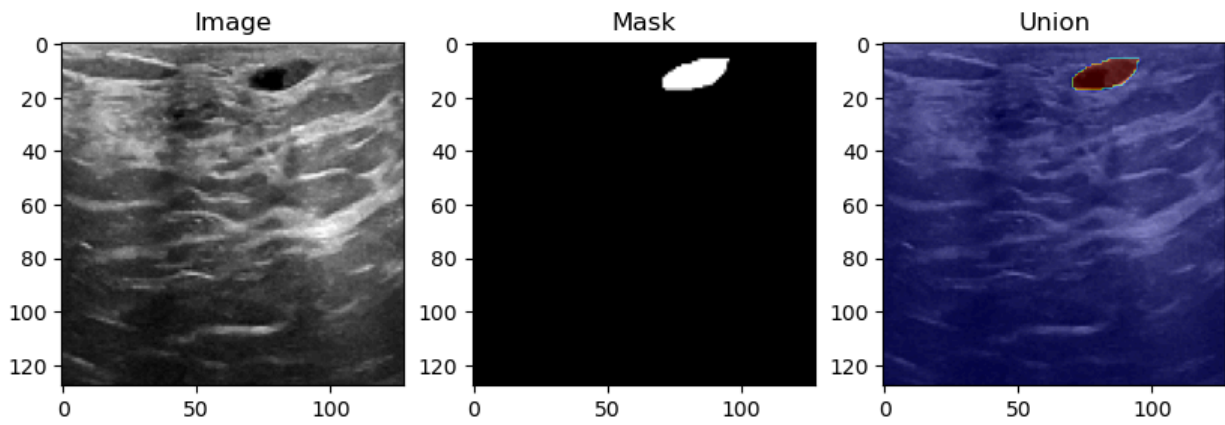
Normal Class

Normal class



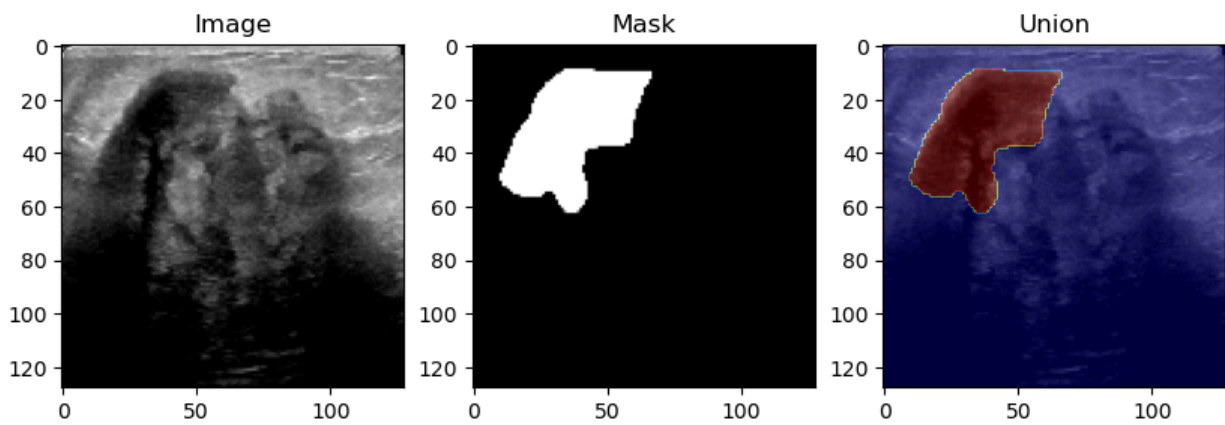
Benign Class

Benign class

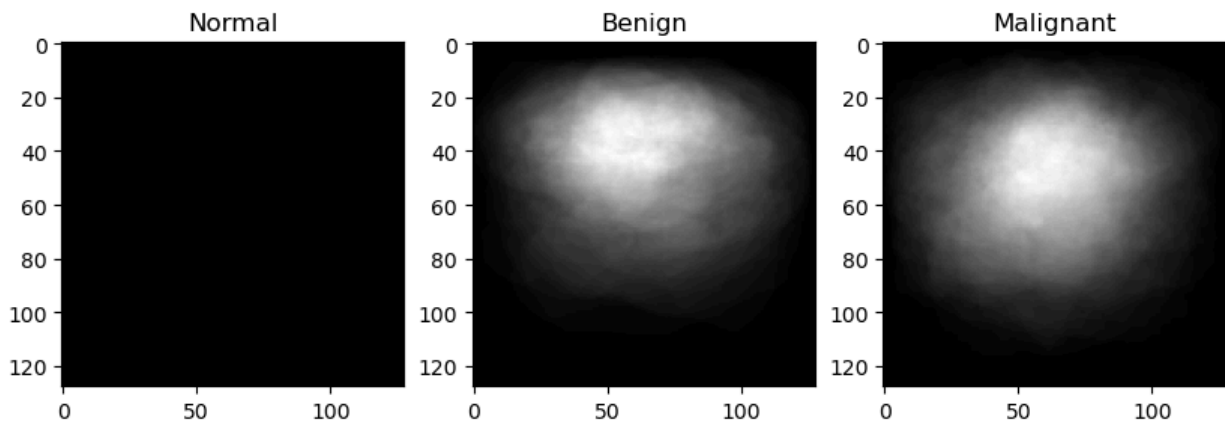


Malignant Class

Malignant class



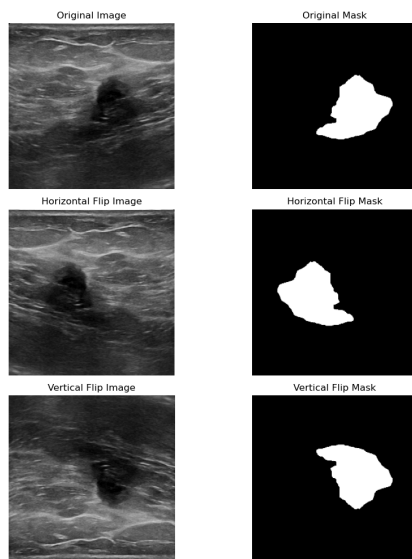
Combination of Places Where there is Cancer:



The image above shows where are the usual regions to find cancer. The black part shows there is no cancer and the white or light part shows there is cancer.

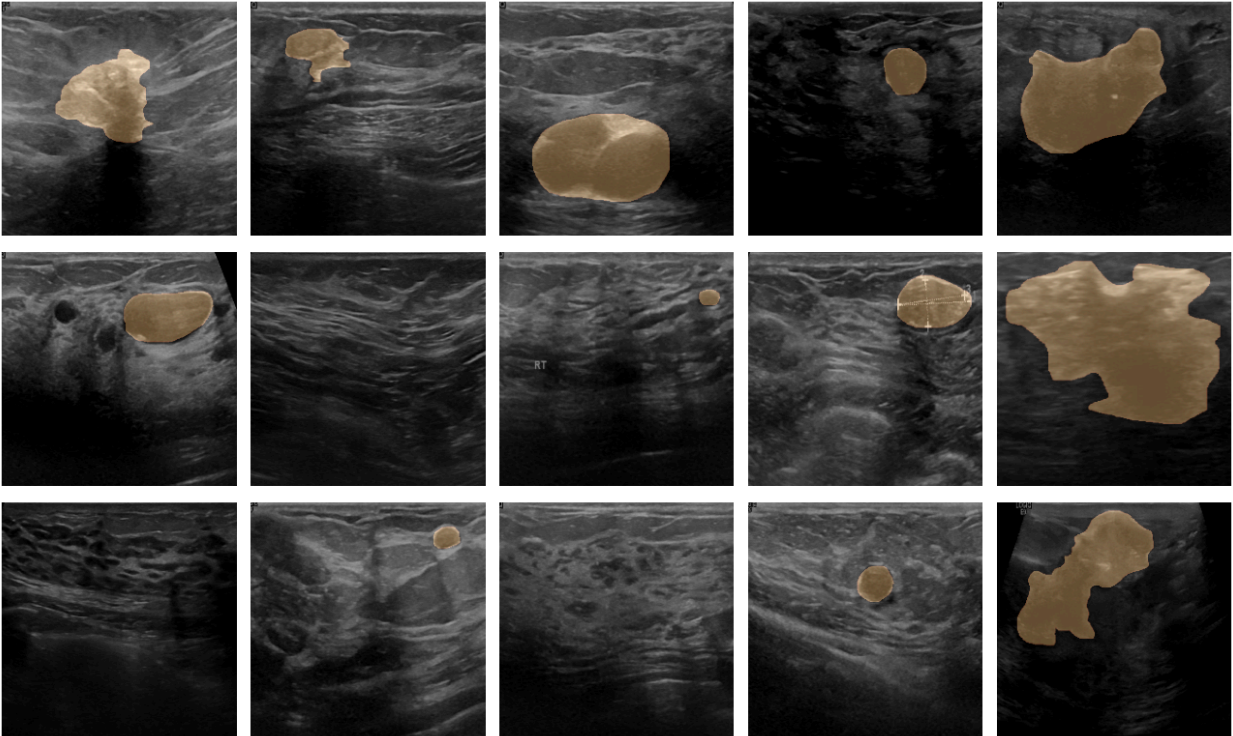
Augmentation

As the number of images are very less, I augmented the data using techniques like Horizontal Flip and Vertical Flip. Doing so we increased the number of Ultrasound images and masked images to 1000 (We could augment to more but due to computational resources we augmented in small numbers). The image below shows an example of augmentation.



Union:

Here the segment images is union with the ultrasound images



Augmented images:

Below Image show the techniques used to augment the dataset.

